

Kubernetes Lab 3 Report

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GitHub Repository: <https://github.com/yi-xian0706/Cloud-Computing>

Lab Location: Lab/Lab3

1 Environment Setup

I installed Minikube and kubectl on Windows. I started the Minikube cluster successfully.

```
PS C:\WINDOWS\system32> minikube start --driver=docker --image-mirror-country=cn
* Microsoft Windows 11 Home 25H2 上的 minikube v1.38.1
* 根据用户配置使用 docker 驱动程序
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* 正在使用镜像存储库 registry.cn-hangzhou.aliyuncs.com/google_containers
* 使用具有 root 权限的 Docker Desktop 驱动程序
* 在集群中 "minikube" 启动节点 "minikube" primary control-plane
* 正在拉取基础镜像 v0.0.50 ...
! minikube was unable to download registry.cn-hangzhou.aliyuncs.com/google_containers/kicbase:v0.0.50, but successfully
downloaded docker.io/kicbase/stable:v0.0.50@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93 as
fallback image
* 创建 docker container (CPU=2, 内存=7600MB) .../
* 正在 Docker 29.2.1 中准备 Kubernetes v1.35.1...
* 配置 bridge CNI (Container Networking Interface) ...
* 正在验证 Kubernetes 组件...
- 正在使用镜像 registry.cn-hangzhou.aliyuncs.com/google_containers/storage-provisioner:v5
* 启用插件: default-storageclass, storage-provisioner
* 完成! kubectl 现在已配置, 默认使用 "minikube" 集群和 "default" 命名空间
PS C:\WINDOWS\system32> kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:55449
CoreDNS is running at https://127.0.0.1:55449/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.
```

2 Deploy Application

I created a deployment named kubernetes-bootcamp using the official tutorial image. The pod started and ran successfully.

```
PS C:\WINDOWS\system32> kubectl create deployment kubernetes-bootcamp --image=registry.aliyuncs.com/google_samples/kuber
netes-bootcamp:v1
deployment.apps/kubernetes-bootcamp created
```

3 Explore Application

I checked the running pods and viewed pod details to ensure the application works correctly.

```
PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
kubernetes-bootcamp-69ff769467-z7978 1/1      Running   0            27s
```

4 Expose Service

I exposed the deployment as a NodePort service. I accessed the application through the browser and saw the correct "Hello Kubernetes bootcamp!" page.

```
PS C:\WINDOWS\system32> kubectl expose deployment kubernetes-bootcamp --type=NodePort --port=8080
service/kubernetes-bootcamp exposed
PS C:\WINDOWS\system32> minikube service kubernetes-bootcamp --url
http://127.0.0.1:55524
! 因为你正在使用 windows 上的 Docker 驱动程序, 所以需要打开终端才能运行它。
```

5 Scale Application

I scaled the application from one replica to three replicas. All replicas are running properly.

```
PS C:\WINDOWS\system32> kubectl scale deployment kubernetes-bootcamp --replicas=3
deployment.apps/kubernetes-bootcamp scaled
PS C:\WINDOWS\system32> kubectl get pods
NAME                                READY    STATUS    RESTARTS    AGE
kubernetes-bootcamp-69ff769467-flqlt 1/1      Running   0            5s
kubernetes-bootcamp-69ff769467-wnwk1 1/1      Running   0            5s
kubernetes-bootcamp-69ff769467-z7978 1/1      Running   0           2m58s
```

6 Update Application

I performed a rolling update from version v1 to v2. The update completed successfully.

```
PS C:\WINDOWS\system32> kubectl set image deployments/kubernetes-bootcamp kubernetes-bootcamp=docker.io/jocatalin/kubernetes-bootcamp:v2
deployment.apps/kubernetes-bootcamp image updated
PS C:\WINDOWS\system32> kubectl rollout status deployments/kubernetes-bootcamp
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
deployment "kubernetes-bootcamp" successfully rolled out
```

7 Clean Up

I deleted the service and deployment, then stopped the Minikube cluster.

Summary

I successfully completed all steps of the Kubernetes basics tutorial using the official kubernetes-bootcamp image. All functions work as required.